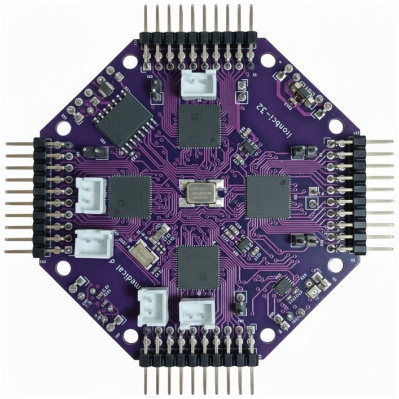
CONNECTOR PINOUT & ELECTRODE GUIDE · SKU PIA24025P

Electrodes, Headers & Pin Order

How to connect 32 electrodes, the reference and the bias to IronBCI-32: the four headers, the top and bottom rows, the non-sequential pin order and the 32-channel 10-20 electrode layout.



4	8	REF	BIAS	10-20
Headers, 2 × 10, 2.54 mm	Channels per header	Top row, pin 1 · CH 1–8	Bottom row, pin 2 · CH 1–8	32-site layout

01 HEADER OVERVIEW

Header	Channels	Converter	Position (user-guide view)	Also carries
CH 1–8	1–8	AD7771 #1	Bottom left	**Reference and bias ear clips**
CH 9–16	9–16	AD7771 #2	Top left	—
CH 17–24	17–24	AD7771 #3	Top right	—
CH 25–32	25–32	AD7771 #4	Bottom right	—

In the user-guide view the **Data Stream** USB port is at the top, between headers 9–16 and 17–24; the **Power** USB port is at the bottom, between headers 1–8 and 25–32.

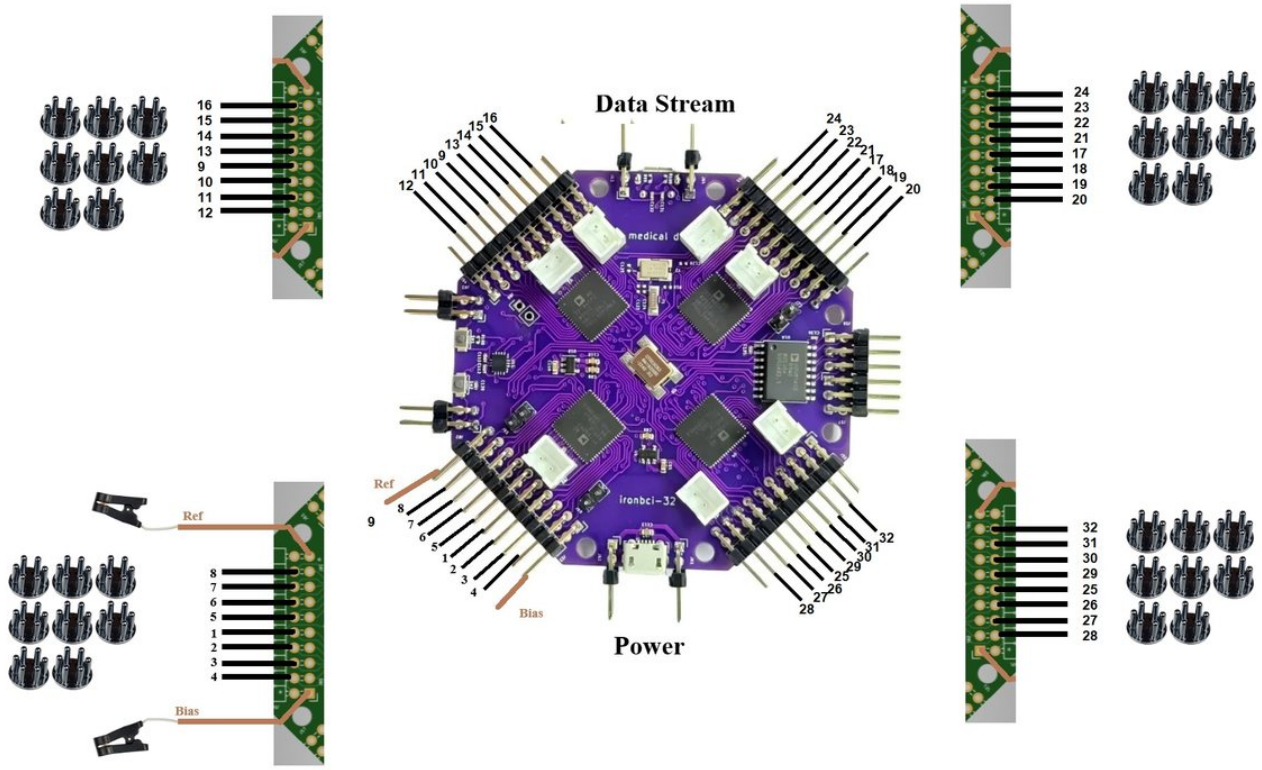


Figure 1 — Wiring overview: four headers with electrodes; reference and bias ear clips on the CH 1–8 header.

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02 TOP AND BOTTOM ROWS

Row	What goes there
Bottom row (closest to the board)	The 8 EEG electrodes, plus **BIAS** on pin 2 (CH 1–8 header)
Top row (further from the board)	Only the **REF** ear clip, on pin 1 (CH 1–8 header). Leave all other top-row pins empty.
Outer end pins (AVSS / AVDD)	Power pins. **Never connect anything.**

WARNING Flat, dead channels?
An electrode plugged into the **top row** is shorted to the reference and shows a flat line. If channels look dead, check the row first.



Figure 2 — Connector diagram: top view, side view (where the rows come from) and front view with channel numbers.

03 PIN ORDER

Looking **into** a connector from outside, with the chips facing up, the pin-1 end (REF / BIAS) is on the **right**. Channels on the bottom row are **not** in simple numerical order. Each cell shows the channel number and its default 10-20 position.

CH 1–8 header of AD7771 #1

TOP row	AVSS do not use	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	REF pin 1
BOTTOM	AVDD do not use	8 CP3	7 P3	6 O1	5 P7	1 F7	2 FT7	3 T7	4 TP7	BIAS pin 2

← outer end (power pins) pin-1 end →

CH 9–16 header of AD7771 #2

TOP row	AVSS do not use	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	REF use CH1–8
BOTTOM	AVDD do not use	16 CPz	15 Cz	14 FCz	13 Fz	9 C3	10 FC3	11 F3	12 Fp1	BIAS use CH1–8

← outer end (power pins) pin-1 end →

CH 17–24 header of AD7771 #3

TOP row	AVSS do not use	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	REF use CH1–8
BOTTOM	AVDD do not use	20 P4	19 O2	18 Oz	17 Pz	21 CP4	22 C4	23 FC4	24 F4	BIAS use CH1–8

← outer end (power pins) pin-1 end →

CH 25–32 header of AD7771 #4

TOP row	AVSS do not use	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	REF use CH1–8
BOTTOM	AVDD do not use	28 T8	27 FT8	26 F8	25 Fp2	29 TP8	30 P8	31 Fpz	32 POz	BIAS use CH1–8

← outer end (power pins) pin-1 end →

Pin order table

Bottom-row pin (left → right)	CH 1–8	CH 9–16	CH 17–24	CH 25–32
1st (power pin)	AVDD	AVDD	AVDD	AVDD
2nd	**8**	**16**	**20**	**28**
3rd	**7**	**15**	**19**	**27**
4th	**6**	**14**	**18**	**26**
5th	**5**	**13**	**17**	**25**
6th	**1**	**9**	**21**	**29**
7th	**2**	**10**	**22**	**30**
8th	**3**	**11**	**23**	**31**
9th	**4**	**12**	**24**	**32**
10th (pin-1 end)	**BIAS**	BIAS*	BIAS*	BIAS*

* Use the bias pin on the CH 1–8 header. On the top row only the pin-1 end (REF) is used; the other top-row pins are internally tied to REF and the leftmost is a power pin (AVSS).

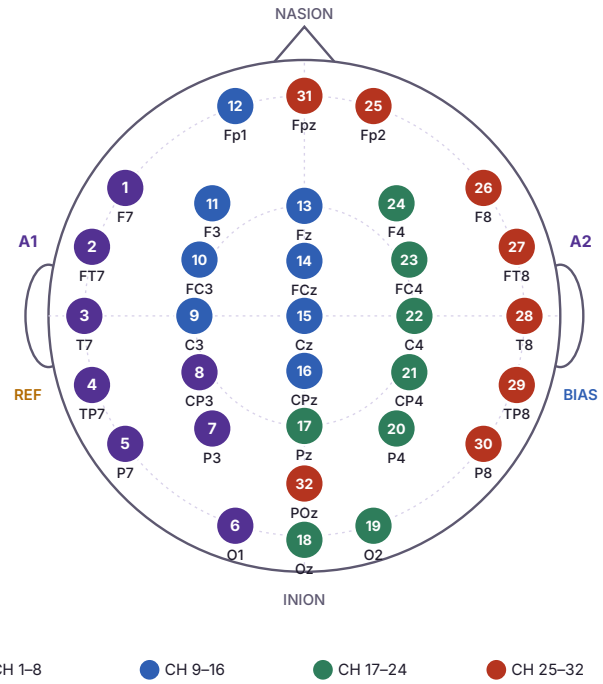
04 REFERENCE AND BIAS

Electrode	Where it goes on the head	Where it plugs in
Reference	Ear clip on one earlobe (A1)	CH 1–8 header, **top row, pin 1**
Bias	Ear clip on the other earlobe (A2)	CH 1–8 header, **bottom row, pin 2**

NOTE The reference and bias clips matter most.

A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly before looking at anything else.

05 ELECTRODE LAYOUT (10-20 SYSTEM)



Ch	Site	Ch	Site	Ch	Site	Ch	Site
1	F7	9	C3	17	Pz	25	Fp2
2	FT7	10	FC3	18	Oz	26	F8
3	T7	11	F3	19	O2	27	FT8
4	TP7	12	Fp1	20	P4	28	T8
5	P7	13	Fz	21	CP4	29	TP8
6	O1	14	FCz	22	C4	30	P8
7	P3	15	Cz	23	FC4	31	Fpz
8	CP3	16	CPz	24	F4	32	POz

Matches the PiEEG 32-channel wet cap. Channels 1–12 cover the left hemisphere, 13–18 the midline and 19–30 the right, with Fpz and POz on 31 and 32. Any other layout can be used — record which channel went where.

06 FITTING THE CAP

Step	Action
1	Measure the head circumference and pick the right cap size.
2	Find **Cz** — halfway between nasion and inion, and halfway between the ears — and align the cap with it.
3	Inject a small amount of conductive gel under each electrode and work it through the hair with the blunt syringe tip.
4	Attach the ear clips to clean earlobes: REF and BIAS.
5	Bundle the 32 leads together and keep them short and still — moving cables show up as noise.

07 RELATED DOCUMENTS

Doc. No.	Document	Contents
IRB32-DS-001	Product Datasheet	Features, specifications, block diagram, ordering
IRB32-MD-002	Mechanical & Dimensional Drawing	Outline, mounting holes, connector positions
IRB32-PCB-003	PCB Technical Information	Stack-up, layers, drill table, fabrication data
IRB32-QSG-005	Quick Start & Safety Guide	Setup, battery-only operation, troubleshooting
IRB32-SW-006	Data Interface & Software Guide	Serial frame, µV conversion, SDKs, firmware

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